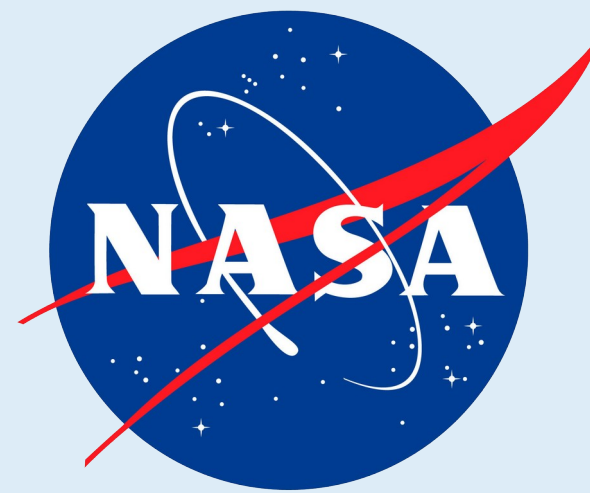




Cross-mission expiMap analysis recovers established tissue responses and identifies complementary pathway shifts in mouse spaceflight transcriptomes

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ABSTRACT

Mission differences can obscure spaceflight responses shared across studies. We mapped NASA OSDR mouse bulk RNA sequencing samples to tissue-matched ARCHS4 references and analyzed about 2,000 highly variable genes with Reactome-constrained expiMap. Flight-ground shifts were balanced across projects and evaluated with gene-set enrichment, project holdouts, three complete training runs, composition proxies, member-gene review, and literature annotation after scoring. Retained programs were lower for thymic repair, skin maintenance, liver adaptive immunity, and splenic adaptive and innate immunity. Spleen showed the strongest result across multiple pathways.

INTRODUCTION

PROJECT OBJECTIVE

Identify gene programs that shift consistently across spaceflight missions.

THE CENTRAL CHALLENGE

Observed expression **Spaceflight biology** + **Tissue biology** + Animal variation + Study / protocol

Study identity and protocol can dominate an unconstrained expression representation. Tissue-matched reference mapping, accession conditioning, and equal project weighting reduce this bias without claiming to remove all mission confounding.

FINAL ANALYSIS SCOPE

Tissue	ARCHS4	OSDR	Projects	Programs
Thymus	1,362	117	5	387
Skin	2,593	151	4	319
Liver	5,000	197	9	364
Spleen	6,289	100	5	360

OSDR counts are primary analysis samples. Spleen excludes one condition-strain-confounded project.

ROBUSTNESS GATE

ssGSEA + GSEA

Held-out projects

3 full trainings

Composition proxies

Member genes

Literature review

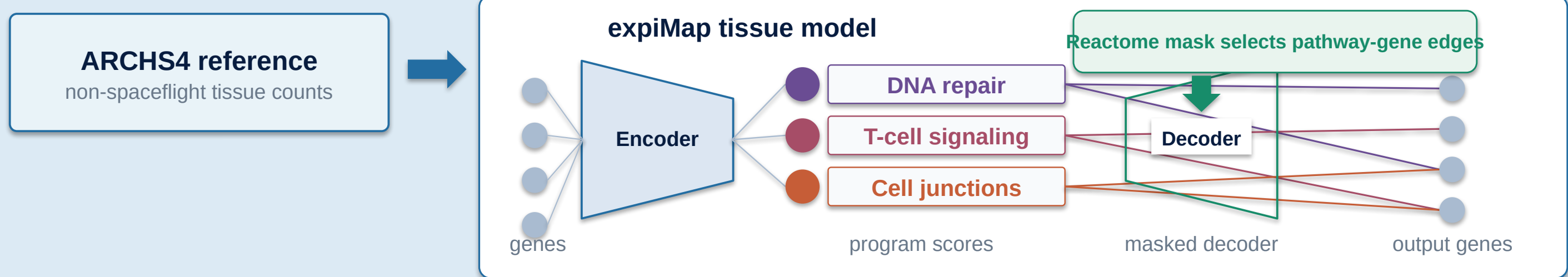
Complementary programs add a plausible perspective beyond the dominant phenotype in prior literature. De novo nodes were not retained in the final models.

METHODS

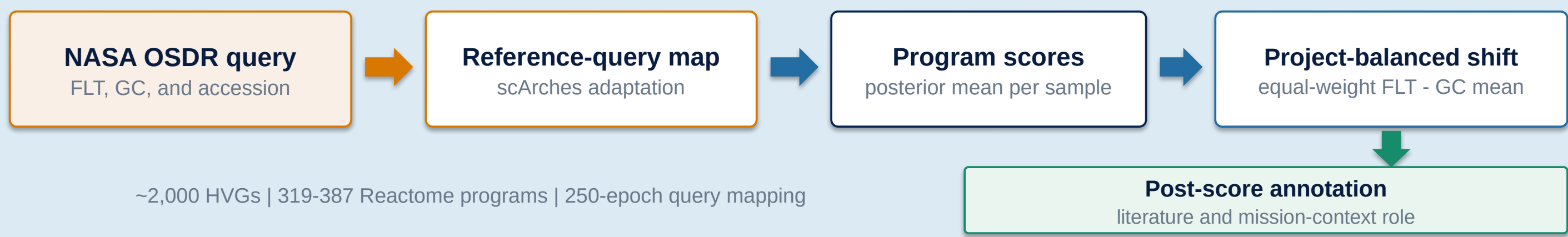
Pathway structure is wired into the decoder

Train a tissue reference, then map flight and ground samples into the same annotated program space.

1 REFERENCE TRAINING

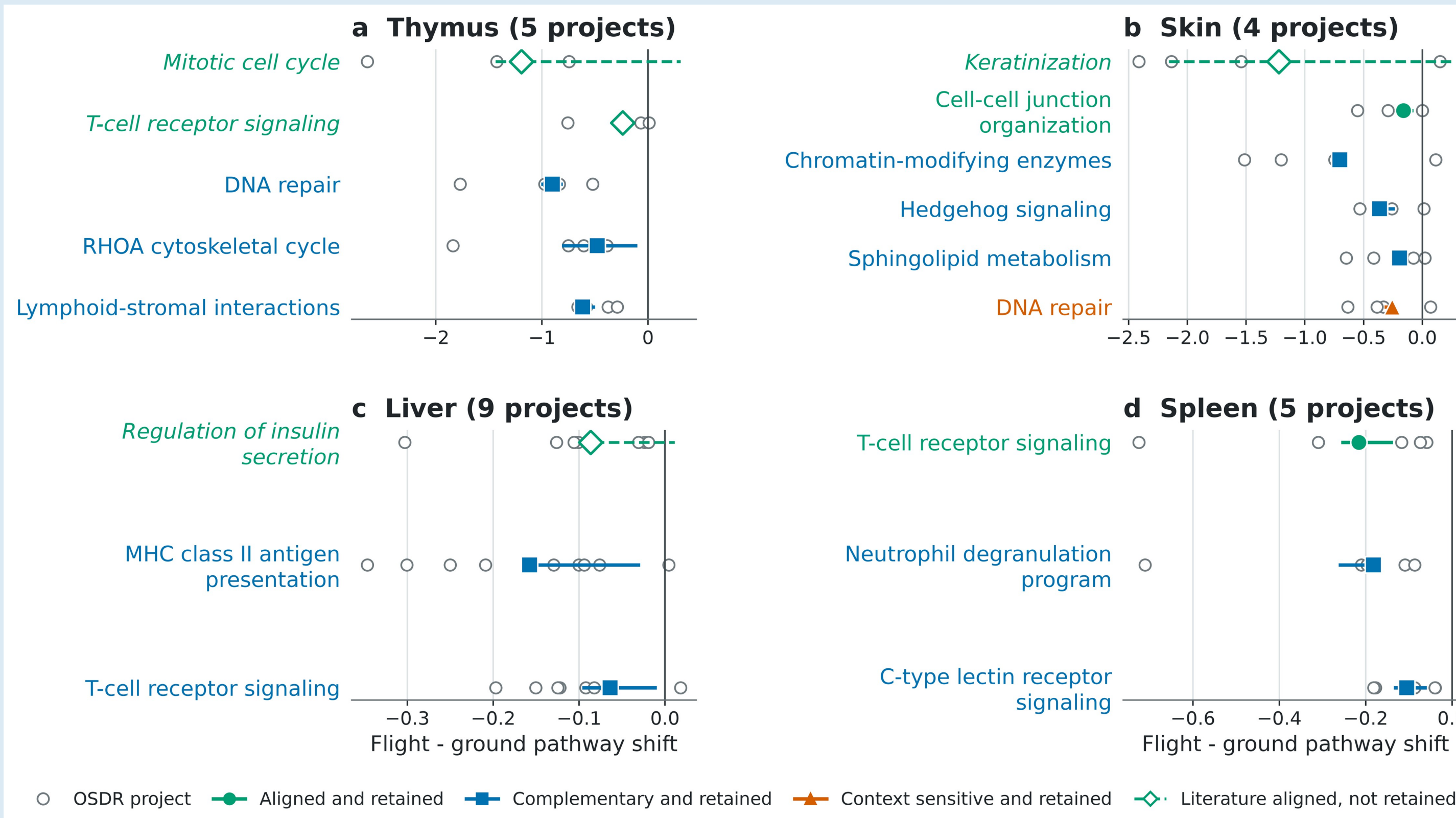


2 QUERY MAPPING AND CROSS-MISSION COMPARISON



~2,000 HVGs | 319-387 Reactome programs | 250-epoch query mapping

RESULTS: ALIGNED ANCHORS AND RETAINED PROGRAM SHIFTS

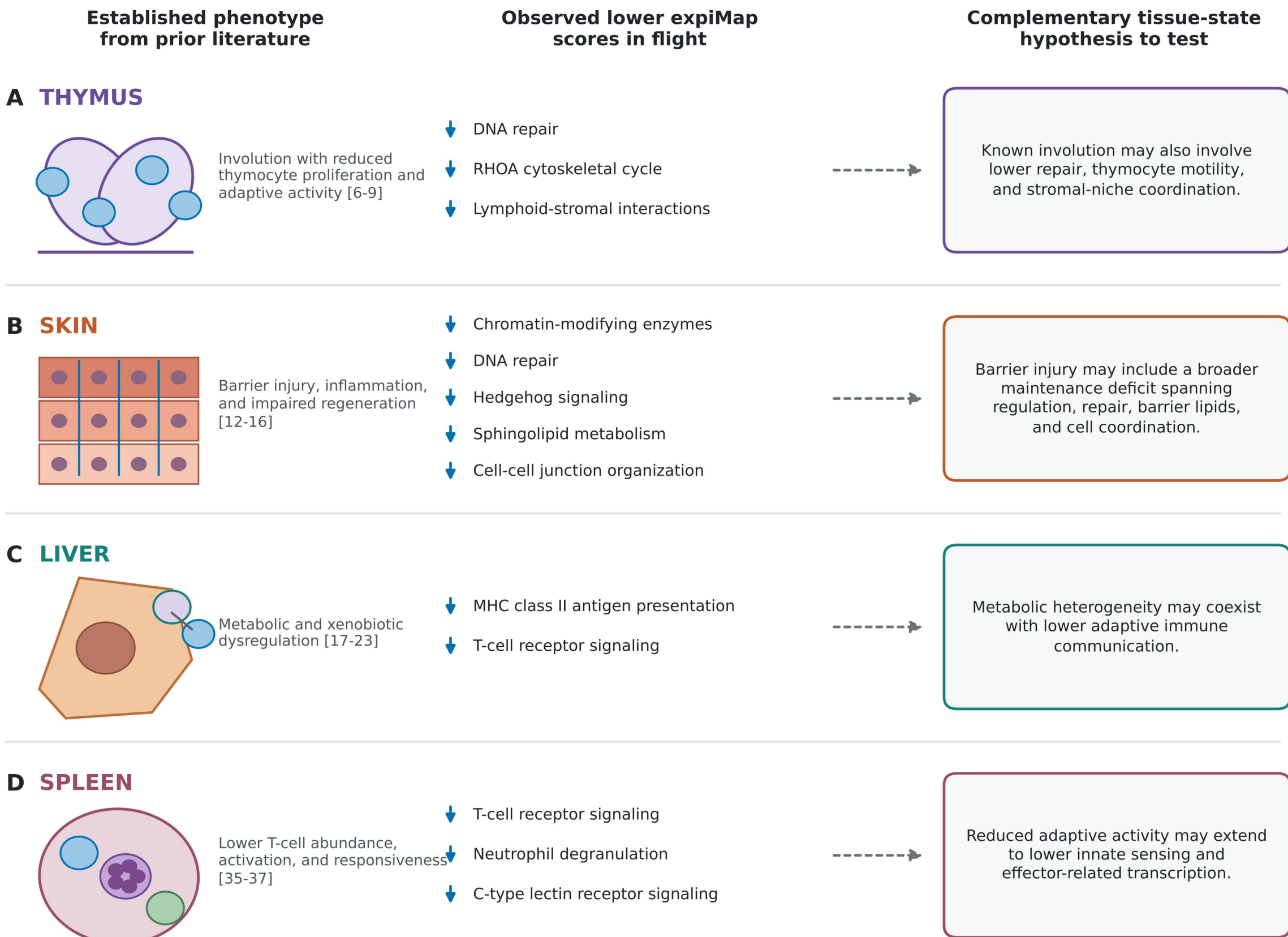


Filled markers show 13 retained programs. Open diamonds are literature-aligned context only, not retained findings; keratinization reversed in one of three trainings. Colored ranges span three complete trainings; axes are tissue specific.

RESULTS: BIOLOGICAL INTERPRETATION

Prior evidence, observed shifts, and hypotheses to test

Prior literature, observed pathway shifts, and hypotheses to test



Prior literature, this study's observations, and new hypotheses are shown separately; dotted arrows denote inference, not causality. Lower scores do not prove pathway inhibition, and cell composition may contribute.

CONCLUSIONS

- Spleen was strongest: T cell receptor, neutrophil degranulation, and C-type lectin receptor programs were lower across five projects and three trainings; all had GSEA FDR <0.05.
- Skin and thymus emphasized lower tissue maintenance programs; liver showed lower adaptive immune signaling amid heterogeneous metabolism.
- Bulk latent scores are testable hypotheses, not causal or cell-resolved measures of pathway activity.

REFERENCES

1 Gebre et al., NAR 2025 (OSDR). 2 Lotfollahi et al., Nat Cell Biol 2023 (expiMap). 3 Lachmann et al., Nat Commun 2018 (ARCHS4). 4 Milacic et al., NAR 2024 (Reactome). 5 Horie et al., Sci Rep 2019. 6 Cope et al., Commun Med 2024. 7 Beheshti et al., Sci Rep 2019. 8 Gridley et al., J Appl Physiol 2009.

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github.com/jasont314/nasa-mouse